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Numerical design and analysis of the tractive performance of large off-road wheels driving on dry sands by DEM

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ABSTRACT: Aimed at the typical condition of large off-road wheels operating on dry sands, in which the sand apparently gets discontinuous and rutted by lugged wheels with larger load, the numerical test for tractive performance is designed and analyzed by discrete element method. It is indicated from the analysis that: Off-road wheels equipped with paralleled lugs show the superiorly tractive performance, especially in aspects of adhesion and tractive force, while V-shaped lugs come second. Types of lugs have little relationship with the sinkage and the efficiency coefficients, which closely reach the maximum in range of 20%-30% of slip. Nonetheless, lugged wheels' performance universally precedes a smooth one's.

Keywords: tractive performance; discrete element method; off-road wheels; dry sand; numerical analysis

1 INTRODUCTION

As the demand of resources increases, human gradually expand the mining areas to the dry and incompact environments of sands, such as desert and gobi in western China and even other planets^[1]. Because of the superior trafficability on sands, off-road vehicles equipped with large lugs play a significant role in resource exploration and transportation. Study of large off-road wheels demands large-scale equipments, which cost so much thus its numerical simulation attracts more attention. Commonly used methods of sands simulation are finite element method (FEM) and discrete element method (DEM). The widely-used FEM depends on the continuous soil model based on the theory of Mohr-Coulomb, Duncan, Bekker, et al.^{[2][3]}, which all hypothesize soil as a continuous medium in advance^[4]. Actually, soil will be deeply rutted by the shearing of operating wheels, especially in case of high slip along with the soil rupturing and splashing in a discontinuous state^[5], which is more apparent on the condition of large off-road wheels with various lugs. In general, FEM exposes a lack of accuracy in analyzing large off-road wheels on dry sands.

No longer simulating the sand as a continuous medium, DEM separates it into a large amount of inde-

pendent particles, which are able to effectively simulate the extrusion and friction among the real sand particles, particularly suitable for numerical analysis of the large off-road wheels with lugs. DEM has been gradually applied in the study of wheeled vehicle on soft ground^[7]. However, a majority of the existing researches face the vehicles under light load like lunar exploration, which may be quite different from the large off-road vehicles in the deformation and destruction of sands resulting from the large size and weight. As the result, analysis of the interaction between large off-road wheels and soft ground demands further consideration.

Aimed at the typical condition of large off-road wheels operating on dry sands, it numerically simulates the sand model by DEM, together with the lugged wheels with the help of a special command called CLUMP in PFC3D. On this basis, calculation and analysis of the tractive parameters with different lugs are completed during the simulation, which reasonably evaluate the performance in different aspects.

2 SIMULATION TEST CONDITIONS

2.1 Conditions of the large off-road wheels

Large off-road wheels' stiffness is always much high-

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er than sand's while driven and operated on it. Hence, the off-road wheel and its lugs are reasonably designed as a rigid model in this paper. The paralleled, V-shaped and inverted V-shaped (from the view of the direction the wheel rolling in) wheel lugs are chosen to be tested in the follow research.

As the increase of the light load in hundreds of times, here come several problems in the large wheel simulation if the method doesn't change. In order to avoid those issues, the models are designed by the particles small enough in PFC3D to make sure the relative smoothness of their surface, which are defined as CLUMP units (shown in Figure 1) together with parameters of these models listed in Table.1.

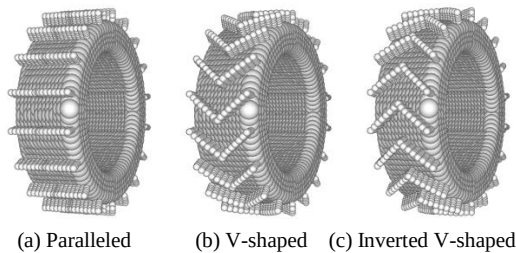


Figure 1. Models of the large off-road wheels

Table 1. Parameters in the model of large off-road wheels

Parameters / units	Assignment
Diameter & width / m	0.94 & 0.32
Vertical load / N	5000
Rotational speed / rpm	5.02
Slip ratio	0, 10, 20, 30, 40, 60, 80
Height of wheel lug / m	0.06
Shape of wheel lug	Paralleled, V-shaped & inverted V-shaped

Note: * Driving speed of wheels is controlled as the change of slip ratio with the constant rotational speed assigned.

2.2 Condition of the tested sands

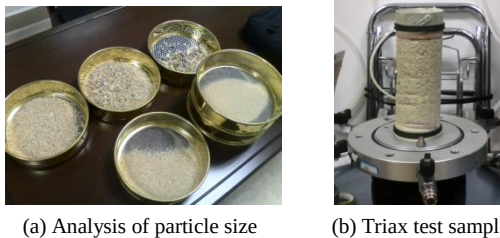


Figure 2. Samples of the simulated sands

The tested sand were sampled from the nearshore zone of a natural bench in Daya Bay, Huizhou, China, to construct a typical operating condition that a large off-road vehicle demands to go through an area of soft and incompact sands. Started with a simple one, the sand with extremely low moisture is selected in this paper as the single condition of the simulation test.

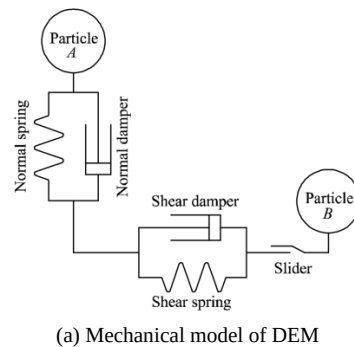
Some necessary tests were carried out before numerical simulation, shown in Figure 2 which contributed to the assignment of specific parameters by DEM in section 2.

3 SIMULATION OF THE SANDS BY DEM

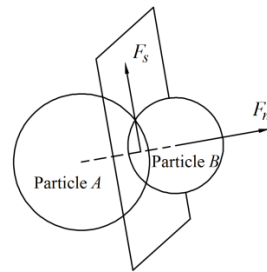
Depending on the theory of DEM, a commercial analysis software PFC3D is applied in this section to establish the discrete element model of the sands, which refers to the macroscopically mechanical properties of the sample for calibration.

3.1 Theory of discrete element method

The theory of DEM simulating the sand is to gather a large amount of spherical particles in a finite volume, in which these particles are stacked and compressed, appearing as an aggregation with the properties similar to the real ones. Any two contiguous particles are connected by the strings, dampers and a slip, shown in Figure 3(a). The string, damper and slip in the mechanical model separately represent the elastic, damping and frictional effects among particles^[1].



(a) Mechanical model of DEM



(b) Contacting condition of particles

Figure 3. Theoretical model of DEM

While there is any normal contact and overlap u between particles, shown in Figure 3(b), normal contact force F_n will be immediately produced and calculated in Equation (1). At that time, any tangentially relative displacement v will cause the tangential contact force F_s due to Equation (2). On account of the slip, there will be relative slipping between the two

contiguous particles, when the tangential contact force F_s no longer increases by its displacement, just as Equation (3):

$$F_n = \eta_n \frac{du}{dt} + K_n u \quad (1)$$

$$F_s = \eta_s \frac{dv}{dt} + K_s v \quad (2)$$

$$F_s \leq \mu F_n \quad (3)$$

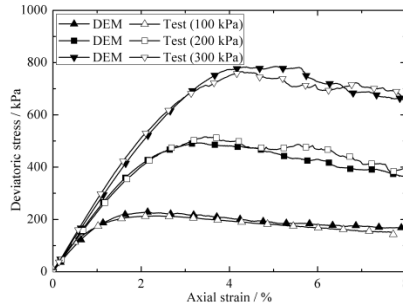
where η_n & η_s are the normal & tangential damping coefficients; K_n & K_s are normal & tangential stiffness factors; μ is the frictional coefficient between particles and u & v are normal & relatively tangential displacement between them.

3.2 Calibration of the sands model by DEM

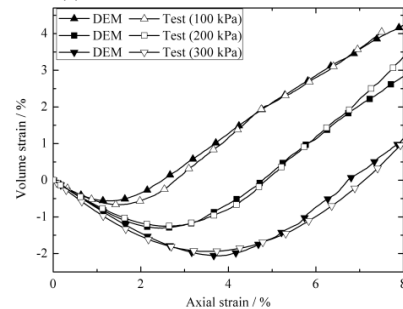
Basis of the design of the sand calculating model is that it shows the same mechanical properties as the real one in triax test, shown in Figure 2(b), which can measure and work out the soil's mechanical parameters by Equation (4) & (5) and plot the trial curves in Figure 4(a) & (b) [8][9]. On that basis, results from the sample of real sands have to be inoculated with the one by DEM. This process is called calibration in DEM.

$$E = \frac{\Delta \sigma}{\Delta \varepsilon} \quad (4)$$

$$v = \frac{1}{2} \left(1 - \frac{\Delta \varepsilon_v}{\Delta \varepsilon} \right) \quad (5)$$



(a) Deviatoric stress vs. axial strain



(b) Volume strain vs. axial strain

Figure 4. Triax test of the soil computing model by DEM

During the incessant attempts on triax test by DEM for the calibration of sands, the parameters of particles in DEM are assigned in Table 2. Comparison of mechanical parameters between the sample and DEM is shown in Table 3, where the errors validate that the DEM model acts in nearly the same mechanical properties with the real sand and the calibration is successful.

Table 2. Parameters of the particles in DEM

Parameters / units	Assignment
Particle size range / mm	18-35
Porosity	0.40
Frictional coefficient	1.23
Normal stiffness / N·m ⁻¹	3.35×10 ⁶
Tangential stiffness / N·m ⁻¹	2.95×10 ⁶
Normal damping	0.7
Tangent damping	0.7

Table3. Comparison of mechanical parameters

Mechanical Parameters	DEM	Sample	Error
Internal cohesion / kPa	3.09	3.31	7.12%
Internal friction angle / °	32.76	34.18	4.33%
Elasticity modulus / MPa	24.47	23.20	5.19%
Poisson's ratio	0.16	0.12	25.0%

Soil bin equipped with the function of measuring force, displacement and sink is an important instrument in the study of terramechanics [10]. Hence, the numerical test is essentially the reappearance of soil bin test by DEM. Depending on the efficient calibration above, 19242 particles are gathered in the volume of 5 m×1 m×0.5 m to simulate the operating conditions of sands with the help of PFC3D, shown in Figure 5. Until now, the numerical test's conditions for tractive performance of off-road wheels have been initially set up.

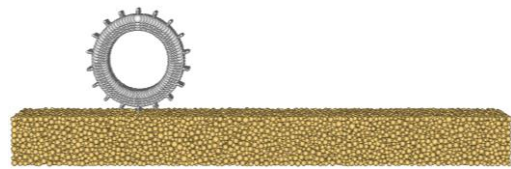


Figure 5. Calibrated sands simulating the soil bin by DEM

4 ANALYSIS OF THE TRACTIVE PERFORMANCE OF LARGE OFF-ROAD WHEELS

Three types of lugged wheels and a smooth one will be tested for the parameters of tractive performance using the DEM analysis software PFC3D. Some differences among the three types of lugged wheels and different tractive parameters will be extracted from the

numerical test, together with the reasonable speculation in micro-analysis.

4.1 Calculation of the tractive parameters by DEM

Another advantage of the method using CLUMP units to simulate the off-road wheels is that the record and calculation of the tractive parameters may be greatly simplified, because the force and displacement of CLUMP units can be directly measured in PFC3D. Tractive parameters in other simulation methods of wheels always need to be complexly extra calculated based on the function designed in advance. Tractive parameters from measurement or calculation in this section are listed in Table 4.

Table 4. Different parameters of tractive performance

Tractive parameters / units	Symbol	Access
Sinkage of the wheel / m	z	measured
Total tractive force	F_{DP}	measured
Moment of torque / N·m	M	measured
Adhesive force from sands / N	F_H	Eq. (6)
Driving resistance from sands / N	F_R	Eq. (7)
Tractive coefficient of efficiency	TE	Eq. (8)

The calculation formulas of the tractive parameters which cannot be measured directly are following [11]. It is proved that F_{DP} equals F_H minus F_R from the test, which verifies the validity of calculation methods, shown in Figure 6 for an example.

$$F_H = \sum_{i=1}^n F_{ball}, \text{ if } F_{ball} > 0 \quad (6)$$

$$F_R = -\sum_{i=1}^n F_{ball}, \text{ if } F_{ball} < 0 \quad (7)$$

$$TE = \frac{F_{DP} \cdot R}{M} (1 - i) \quad (8)$$

where F_{ball} represents the force acted on each particle of the wheel in the CLUMP unit; R is the radius of the wheel regardless of its lugs and i represents the slip ratio.

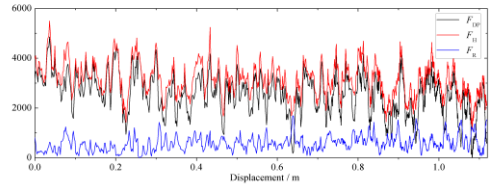


Figure 6. Records of F_{DP} , F_H & F_R of paralleled lugs ($i=80\%$)

4.2 Results and analysis from the numerical test

Depending on the methods of measurement and calculation of the specified tractive parameters above, the relation between those parameters and slip ratio with different off-road lugs are plotted in Figure 7. Figure 7(a) infers that the sinkage of wheels approximately increases by the slip ratio, which is more apparent during the period of low slip, and has little relation-

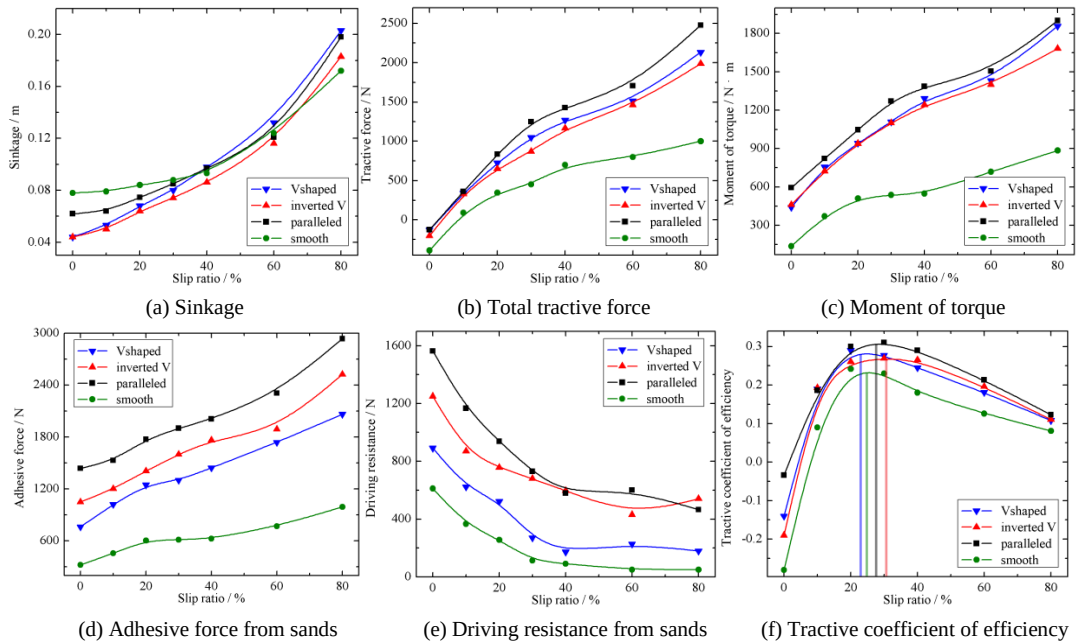


Figure 7. Tractive parameters of large off-road wheel versus slip ratio

ship with the types of wheel lugs.

Figure 7(b) & (c) indicate that both total tractive force and moment of torque the wheel received from sands rapidly increase at the very beginning of slip. The slopes gradually decrease as the slip increases, which match Bekker's semi-empirically theoretical model widely accepted already^[2]. However, the difference from the Bekker's model is that those slopes of the simulation in this paper increase again when slip reaching up to a high ratio of 60%-80%. This abnormal phenomenon had also been found in numerical simulation and soil bin test by Wen Li & Peng Sun (2008-2010)^{[12][13]}. It may be speculated that the abnormal increases of tractive force and moment of torque are attributed to the relatively high load on large off-road wheels in this paper, while the soil tests in reality usually are based on light weight because of the limit of testing instruments.

In addition, it can be found that wheel lugs are able to significantly improve the tractive performance. The wheel equipped with paralleled lugs shows the superiorly tractive performance, while the lugs of V-shaped reveal a small amount of advantage compared with the one of inverted V-shaped. Besides, the types of lugs hardly make differences in the tractive force and torque at the beginning of slip.

Throughout the calculation of function presupposed in PFC3D, Figure 7(d) & (e) show the adhesive and resistant force from sands, which are difficult to be directly measured in real soil bin test. Figure 7(d) indicates that the adhesive force provided by sand increases by the slip, corresponding to the change of total tractive force and torque, due to the sands' shearing property. Figure 7(e) infers that as the increase of the slip ratio with a constant rotational speed assigned in advance, the resistant force from sands is likely to decrease rapidly at the beginning of slip, approaching to a minimum at a high slip ratio. It can also be speculated that in terms of types of wheel lugs, the larger the adhesion is, the more resistance there will be.

It has been concluded that the slip ratio will significantly contribute to the tractive force and torque of wheel. However, high ratio of slip together with larger tractive force and torque do not make for the high efficiency of trafficability. Figure 7(f) briefly points out that tractive coefficient of efficiency will sharply rise at the very beginning of curve, after reaching a maximum, gradually decrease by the continuing growth in slip. Figure 7(f) also shows that results from three types of wheel lugs are quite close, among which the paralleled lug's is little higher, but apparently higher than the smooth one. Meanwhile, the slips where each coefficient of efficiency reaches the maximum all closely appear in the range of 20%-30% of ratio.

4.3 Micro-analysis by DEM

Simulating the sands as discrete and discontinuous particles from a mesoscopic angle, which may restore the mechanical properties of real sands at the great extent, is the most advantage of DEM. Figure 8 shows the velocity and contact force at different slip ratio of wheels equipped with V-shaped lugs at different time, in which the color more close to red means the higher speed of the particles, and the width of the black lines among particles represents the magnitude of contact force.

It can be inferred that the velocity gets higher by the increase of slip ratio, and the primary velocity appears below the bottom of the wheel, which can be obviously observed in Figure 8(g) at a high slip. Figures of contact force illustrate that the force mainly appears on the cambered surface of the front of wheel contacting the sands, which matches the semi-empirical hypothesis in Bekker's model. Meanwhile, sands are more apparently rutted as the slip increasing, contributing to the severer disturbance of its surface in a discontinuous state. Hence, the micro-analysis confirms the rationality of DEM due to its discrete simulation method of soil, which is more suitable for the incom-

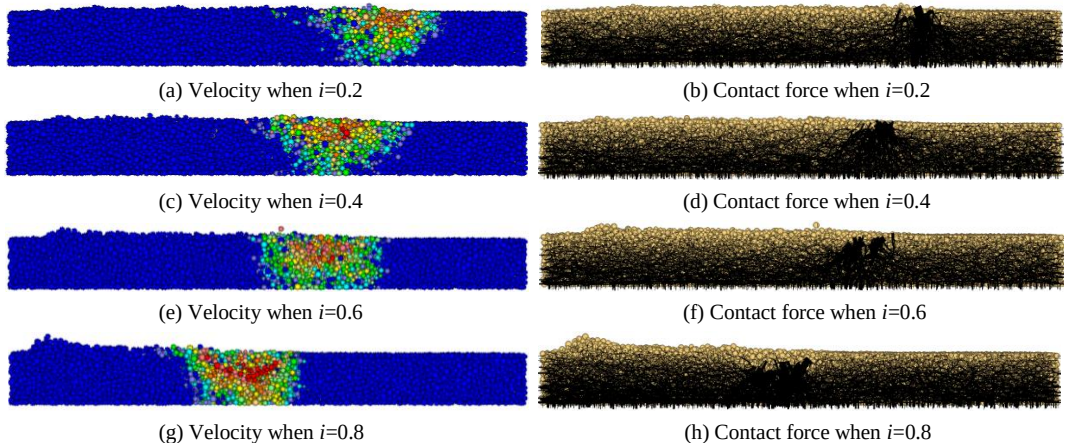


Figure 8. Velocity and contact force at different slip ratios of wheels with V-shaped lugs

pact samples with low moisture like sand.

5 CONCLUSION

Aimed at the condition of large off-road wheels operating on the dry and incompact sands without moisture, three types of lugged wheels are designed by CLUMP units in PFC3D to research their tractive performance on the simulated sands based on DEM theory. Following conclusions have been obtained from the analysis of the simulation test:

1) Slip ratio significantly contributes to the tractive performance in terms of torque and adhesion, and the reduction of resistance from sands under the condition of a constant rotational speed presupposed.

2) Off-road wheel equipped with paralleled lugs shows the superiorly tractive performance on sands, especially in aspects of adhesion and tractive force.

3) Types of wheel lugs have little relationship with the sinkage and the efficiency coefficients, which closely reach the maximum in the range of 20%-30% of slip. Nonetheless, lugged wheels' tractive performance universally precedes a smooth one's.

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